Robust Small–Prime Freezing at the Four–Möbius Gate: Walsh–Face Compression, Exact Prime Descent, and the High–Prime Drift Barrier

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Abstract

TPC–43 represented the surviving equality energy at the four–Möbius gate as a nonnegative Walsh polynomial whose all–minus prime corner is the physical Möbius value. Its annealed theorem controlled almost every prime–sign environment but did not select that corner. We prove a partial de–randomization. Split every squarefree row label at a cutoff z, and let ν_z be the number of distinct small–prime parts occurring in the actual packet. A Hilbert–space grouping and one Cauchy–Schwarz inequality give

$$\mathbb{P}_{>z}\left(\sup_{\sigma_{p\le z}} Z_0(\sigma,\varepsilon_{>z}) > L\nu_z\mathcal{D}_0\right) \le L^{-1}.$$

Thus one high–prime environment controls every small–prime face simultaneously. At the inherited endpoint, a Rankin bound for the projected labels gives $\nu_z \le X^{62549/26302500+o(1)}$ for $z = (\log X)^{501/500}$, strictly below the available $K_B = X^{1/400+o(1)}$ budget. Consequently a density $1-o(1)$ set of high–prime environments remains admissible after all signs $p \le z$ are overwritten arbitrarily, in particular by the physical all–minus values. We also give two exact prime–by–prime transports. The coefficient restriction energy satisfies

$$|Z_0(-\mathbf{1}) - \mathcal{D}_0|^2 = \mathrm{Var}(Z_0)\prod_p(1-\theta_p), \qquad -1 \le \theta_p \le 1,$$

while conditional face energies give an exact likelihood product in an energy–biased measure. A degree–two, entirely z-rough Gram example has vanishing relative variance, vanishing normalized influences, and vanishing prime–toggle quadratic variation, yet its all–minus amplification diverges. Hence variance, fixed moments, rough support, or small local defects cannot finish the deterministic endpoint. What remains is a coefficient–specific high–prime cumulative correlation estimate, naturally of bilinear/Type–II type; no parity breach or prime–pair theorem is claimed.

Keywords: Möbius correlation; Walsh face; random multiplicative function; small–prime freezing; reverse martingale; coefficient descent; sieve parity.

Convention. All prime sets and Walsh polynomials at a fixed scale are finite. Empty coefficient energies are handled by continuity, so normalized correlations are defined only when their denominators are positive.

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1 Introduction and main results

The terminal obstruction isolated in TPC–42 is a coherent Hilbert energy with one source and one target Möbius value in each row [13]. TPC–43 replaced μ by one globally consistent Rademacher multiplicative field and proved exact annealed diagonalization at the equality alias [14]. The same paper compressed every unequal row pair into a squarefree Walsh kernel:

$$Z_0(\varepsilon) = \mathcal{D}_0 + \sum_{\kappa>1} C_X(\kappa)\chi_\kappa(\varepsilon), \qquad Z_0(-\mathbf{1}) = \mathcal{D}_0 + \sum_{\kappa>1} \mu(\kappa) C_X(\kappa). \tag{1}$$

Here $Z_0 \geq 0$, $\mathbb{E}Z_0 = \mathcal{D}_0$, and $Z_0(-\mathbf{1})$ is the literal physical equality energy. The subscript X is suppressed below.

The difficulty is now extremely specific. Annealed estimates describe most vertices of the prime cube, whereas the arithmetic problem asks for one prescribed vertex. A small exceptional set may contain that vertex at every scale. The present paper does not try to hide this quantifier gap. Instead it asks how many prime coordinates can be made adversarial before the annealed endpoint is lost.

1.1 Uniform control of an entire small–prime face

Every nonzero equality atom has a squarefree label u. Split it at $z \geq 2$ as

$$u = u_{\leq z} u_{>z}, \qquad P^+(u_{\leq z}) \leq z < P^-(u_{>z}), \tag{2}$$

with the usual conventions for the factor 1. Let

$$\nu_z := \#\{u_{\leq z} : u \text{ occurs as an equality label}\}. \tag{3}$$

This is an actual packet complexity, not the cardinality $2^{\pi(z)}$ of the entire small–prime cube.

The first main theorem is the following robust face estimate; its abstract version is proved in section 4.

Theorem 1.1 (Robust small–prime face)**.** *For every $z \geq 2$ and $L \geq 1$,*

$$\mathbb{P}_{\varepsilon_p : p>z}\left(\sup_{\sigma \in \{-1,1\}^{\{p \leq z\}}} Z_0(\sigma, \varepsilon_{>z}) > L\nu_z \mathcal{D}_0 \right) \leq L^{-1}. \tag{4}$$

In particular, some high–prime environment satisfies

$$\sup_\sigma Z_0(\sigma, \varepsilon_{>z}) \leq \nu_z \mathcal{D}_0. \tag{5}$$

The supremum is inside the probability. Thus the theorem first selects one high–prime environment and then allows every small–prime sign to be changed, jointly and adversarially. It is stronger than proving a tail bound for each fixed small–prime assignment separately.

At the TPC endpoint the labels obey

$$u \leq X^{\lambda_0 + o(1)}, \qquad \lambda_0 = 1 + \frac{10049}{52500} = \frac{62549}{52500}. \tag{6}$$

A Rankin count of possible projected labels yields the quantitative advance.

Theorem 1.2 (A mildly superlogarithmic frozen block)**.** *Put*

$$z_X = (\log X)^{501/500}, \qquad L_X = \exp\left(\frac{\log X}{\sqrt{\log\log X}} \right). \tag{7}$$

Then

$$\nu_{z_X} \le X^{62549/26302500+o(1)} \tag{8}$$

and, for all sufficiently large X,

$$L_X \nu_{z_X} \le K_B, \qquad K_B = X^{1/400+o(1)}. \tag{9}$$

Consequently

$$\mathbb{P}_{>z_X}\left(\sup_{\sigma_{p\le z_X}} Z_0(\sigma, \varepsilon_{>z_X}) > K_B \mathcal{D}_0\right) \le \exp\left(-\frac{\log X}{\sqrt{\log\log X}}\right). \tag{10}$$

In particular, the signs at all primes $p \le z_X$ may be overwritten by -1, exactly as required by μ. The theorem does not set the remaining high–prime signs to -1.

1.2 Exact descents and the remaining gate

Let $F = Z_0 - \mathcal{D}_0$, extend $C(\kappa)$ by zero, and list the active primes as $p_1 < \cdots < p_R$. For squarefree m, define

$$C_r(m) := \sum_{a \mid p_1 \cdots p_r} \mu(a) C(am), \qquad (m, p_1 \cdots p_r) = 1. \tag{11}$$

Then setting the first r signs to -1 is exactly

$$F(-\mathbf{1}_{\le p_r}, \varepsilon_{>p_r}) = \sum_{(m, p_1 \cdots p_r)=1} C_r(m) \chi_m(\varepsilon). \tag{12}$$

If E_r is the sum of $|C_r(m)|^2$ over squarefree m coprime to $p_1 \cdots p_r$, coefficient pairing gives

$$E_r = (1-\theta_r) E_{r-1}, \qquad -1 \le \theta_r \le 1, \tag{13}$$

and hence the exact identity

$$\boxed{|F(-\mathbf{1})|^2 = \mathrm{Var}(Z_0) \prod_{r=1}^{R} (1-\theta_r).} \tag{14}$$

This is a restriction identity, not a probabilistic estimate.

There is also a nonnegative face transport. Average the first r prime signs while keeping every later one equal to -1, and call the result A_r. When $Z_0(-\mathbf{1}) > 0$,

$$\frac{Z_0(-\mathbf{1})}{\mathcal{D}_0} = \prod_{r=1}^{R} (1+\delta_r), \qquad -1 \le \delta_r \le 1. \tag{15}$$

Under the energy–biased probability $d\mathbb{Q}_Z = (Z_0/\mathcal{D}_0) dU$, each factor is a conditional likelihood:

$$1 + \delta_r = 2\mathbb{Q}_Z\big(\varepsilon_{p_r} = -1 \mid \varepsilon_{p_{r+1}} = \cdots = \varepsilon_{p_R} = -1\big). \tag{16}$$

If $Z_0(-\mathbf{1}) = 0$, the desired upper endpoint is automatic. Thus the nontrivial endpoint selection is exactly a cumulative boundary–path problem.

The distinction between local and cumulative control is essential. In section 8 we construct a degree–two Gram energy with all nonconstant kernels z-rough for which

$$\frac{\mathrm{Var}\, Z}{(\mathbb{E}Z)^2} \longrightarrow 0, \qquad \max_r |\delta_r| \longrightarrow 0, \qquad \sum_r \delta_r^2 \longrightarrow 0, \tag{17}$$

but $Z(-\mathbf{1})/\mathbb{E}Z \to \infty$. Therefore neither a second moment nor small primewise defects can finish the physical corner. The remaining positive target is a one–sided cumulative estimate for the actual $C_X(\kappa)$, after the small primes have been folded exactly.

1.3 Claim boundary and organization

The paper proves a partial de–randomization: inside a density–one set of high–prime environments, a growing mildly superlogarithmic block can be overwritten by the physical signs. It does not prove that the all–minus high–prime environment is typical, that the product in (14) is $X^{o(1)}$, or that the rough tail cancels. In particular, it proves no fixed–shift Chowla estimate, no breach of the sieve parity barrier, no Hardy–Littlewood prime–pair asymptotic, and no infinitude of twin primes.

Section 2 records the inherited interface. Section 3 gives the exact Walsh martingales. Sections 4 and 5 prove the uniform face theorem and its endpoint exponent. The two exact descents are developed in sections 6 and 7. Section 8 gives the sharp obstruction. Sections 9 and 10 compare the result with the literature and state the next coefficient–specific gate. The finite certificate is described in section 11.

2 The inherited equality–Walsh interface

We isolate exactly the part of TPC–43 used here. The left and right output coordinates are combined into one finite index set i. For each i,

$$S_i(\varepsilon) = \sum_{u \in \mathcal{U}_i} b_{i,u}\chi_u(\varepsilon), \qquad Z_0(\varepsilon) = \sum_i |S_i(\varepsilon)|^2, \tag{18}$$

where every u is squarefree and the nonzero labels in a fixed $\mathcal{U}_i$ are pairwise distinct. The characters are

$$\chi_u(\varepsilon) = \prod_{p|u} \varepsilon_p. \tag{19}$$

All coefficients and label sets are deterministic after the inherited TPC packet has been fixed.

The atomic diagonal is

$$\mathcal{D}_0 := \sum_i \sum_{u \in \mathcal{U}_i} |b_{i,u}|^2. \tag{20}$$

Pairwise label distinctness within each coordinate and Walsh orthogonality give

$$\mathbb{E}Z_0 = \mathcal{D}_0. \tag{21}$$

If $\mathcal{D}_0 = 0$, every coefficient vanishes and all statements below are trivial. We henceforth suppose $\mathcal{D}_0 > 0$.

Expanding the squared norms and grouping equal squarefree symmetric–difference kernels gives real coefficients $C(\kappa)$ such that

$$Z_0(\varepsilon) = \mathcal{D}_0 + F(\varepsilon), \tag{22}$$

$$F(\varepsilon) = \sum_{\kappa > 1} C(\kappa)\chi_\kappa(\varepsilon), \tag{23}$$

$$Z_0(-\mathbf{1}) = \mathcal{D}_0 + \sum_{\kappa > 1} \mu(\kappa)C(\kappa), \tag{24}$$

$$V := \|F\|_2^2 = \mathrm{Var}(Z_0) = \sum_{\kappa > 1} |C(\kappa)|^2. \tag{25}$$

Every prime in these formulas belongs to the finite support of the labels. We extend C by zero to all other squarefree integers and set $C(1) = 0$.

At the physical equality alias the labels have the more specific form

$$u_{\alpha,j} = d_\alpha n_{\alpha,j,0} = d_\alpha(\ell_\alpha d_\alpha j + h), \tag{26}$$

where $h \neq 0$ is fixed,

$$\begin{aligned} \ell_\alpha d_\alpha &\asymp Q, & Q &= X^{267/400+o(1)}, \\ j &\asymp J, & J &= X^{133/400+o(1)}, \\ d_\alpha &\asymp D_0, & D_0 &= X^{10049/52500+o(1)}. \end{aligned} \tag{27}$$

The notation D_0 for the divisor scale is distinct from the calligraphic energy $\mathcal{D}_0$. Since $QJ \asymp X$,

$$u_{\alpha,j} \leq X^{\lambda_0+o(1)}, \qquad \lambda_0 = 1 + \frac{10049}{52500} = \frac{62549}{52500}. \tag{28}$$

The equality endpoint budget and inherited diagonal estimate are

$$K_B = X^{1/400+o(1)}, \qquad \mathcal{D}_0 \ll X^{o(1)} Q^2 J, \tag{29}$$

and the desired physical conclusion would be

$$Z_0(-\mathbf{1}) \ll X^{o(1)} K_B Q^2 J. \tag{30}$$

For later comparison, Boolean hypercontractivity [2, 3, 8] gives a soft variance bound. Put

$$d_X := \max_{i, u \in \mathcal{U}_i} \omega(u). \tag{31}$$

The maximal order of ω and (28) imply

$$d_X \leq (\lambda_0 + o(1)) \frac{\log X}{\log\log X}. \tag{32}$$

Applying the L^4 inequality to each S_i, then Minkowski to $Z_0 = \sum_i |S_i|^2$, yields

$$V^{1/2} \leq \|Z_0\|_2 \leq 3^{d_X} \mathcal{D}_0 = X^{o(1)} \mathcal{D}_0. \tag{33}$$

This estimate is useful after a structural compression; by itself it does not evaluate F at one prescribed corner.

3 Forward and reverse prime martingales

List the primes that occur in some nonzero Walsh coefficient as

$$p_1 < p_2 < \cdots < p_R. \tag{34}$$

For endpoint conventions, put $p_0 := 1$; thus the set of prime coordinates at most p_0 is empty. For any cutoff z, write

$$r(z) := \max\bigl(\{r : p_r \leq z\} \cup \{0\}\bigr). \tag{35}$$

The finite cube has two natural filtrations. Although the identities in this section are elementary, they distinguish averaging a coordinate from physically restricting it to -1.

For $1 \leq r \leq R$, the forward filtration is $\mathscr{F}_r^+ = \sigma(\varepsilon_{p_1}, \ldots, \varepsilon_{p_r})$, put

$$M_r := \mathbb{E}(F \mid \mathscr{F}_r^+). \tag{36}$$

For $0 \leq r < R$, the reverse filtration retains the signs strictly larger than p_r:

$$R_r := \mathbb{E}\bigl(F \mid \varepsilon_{p_{r+1}}, \ldots, \varepsilon_{p_R}\bigr). \tag{37}$$

Set $M_0 = R_R = 0$; then $M_R = R_0 = F$.

Proposition 3.1 (Exact largest– and least–prime layers)**.** *For $0 \le r \le R$, the formulas for M_r and R_r below hold; for $1 \le r \le R$, so do the difference formulas:*

$$M_r = \sum_{\substack{\kappa>1 \\ P^+(\kappa)\le p_r}} C(\kappa)\chi_\kappa, \tag{38}$$

$$d_r^+ := M_r - M_{r-1} = \sum_{P^+(\kappa)=p_r} C(\kappa)\chi_\kappa, \tag{39}$$

$$R_r = \sum_{\substack{\kappa>1 \\ P^-(\kappa)>p_r}} C(\kappa)\chi_\kappa, \tag{40}$$

$$d_r^- := R_{r-1} - R_r = \sum_{P^-(\kappa)=p_r} C(\kappa)\chi_\kappa. \tag{41}$$

Moreover,

$$\|d_r^+\|_2^2 = \sum_{P^+(\kappa)=p_r} |C(\kappa)|^2, \tag{42}$$

$$\|d_r^-\|_2^2 = \sum_{P^-(\kappa)=p_r} |C(\kappa)|^2, \tag{43}$$

$$\sum_{r=1}^{R} \|d_r^+\|_2^2 = \sum_{r=1}^{R} \|d_r^-\|_2^2 = V. \tag{44}$$

Proof. A Walsh character survives a conditional expectation exactly when it uses only retained coordinates. This gives the four spectral formulas. The layers have disjoint Walsh support, so orthogonality gives the energy identities. □

Evaluating the reverse martingale on the retained all–minus face gives the filtration from TPC–43:

$$\mathcal{D}_0 + R_r(-\mathbf{1}) = \mathcal{D}_0 + \sum_{\substack{\kappa>1 \\ P^-(\kappa)>p_r}} \mu(\kappa)C(\kappa). \tag{45}$$

Its exact L^2 dissipation is

$$\|R_{r-1}\|_2^2 - \|R_r\|_2^2 = \|d_r^-\|_2^2. \tag{46}$$

There is a deterministic but generally costly layer estimate. Write

$$e_r^- := \sum_{P^-(\kappa)=p_r} |C(\kappa)|^2, \qquad N_r^- := \#\{\kappa : C(\kappa) \neq 0,\ P^-(\kappa) = p_r\}. \tag{47}$$

Then

$$|F(-\mathbf{1})| \le \sum_{r=1}^{R} \sqrt{N_r^- e_r^-}. \tag{48}$$

More generally, for $w_r > 0$,

$$|F(-\mathbf{1})|^2 \le \left(\sum_r \frac{N_r^-}{w_r}\right)\left(\sum_r w_r e_r^-\right). \tag{49}$$

These inequalities are exact diagnostics, not endpoint estimates: the support factors N_r^- can be too large. The next section avoids the full kernel support by returning to the row labels before squaring.

4 Uniform control of a whole small–prime face

Fix $z \geq 2$. Every squarefree equality label has the unique split

$$u = ab, \qquad P^{+}(a) \leq z < P^{-}(b), \tag{50}$$

where $a = 1$ or $b = 1$ is allowed. Let

$$\mathcal{A}_z := \{a : a \text{ occurs as the low part of an actual label}\}, \qquad \nu_z := |\mathcal{A}_z|. \tag{51}$$

For a high–prime environment $\eta = (\varepsilon_p)_{p>z}$, group the coordinate coefficients by their low parts:

$$B_{i,a}(\eta) := \sum_{\substack{u\in\mathcal{U}_i \\ u_{\leq z}=a}} b_{i,u}\chi_{u_{>z}}(\eta), \qquad W_z(\eta) := \sum_i \sum_{a\in\mathcal{A}_z} |B_{i,a}(\eta)|^2. \tag{52}$$

Lemma 4.1 (Projected diagonal identity)**.** *One has*

$$\mathbb{E}_\eta W_z(\eta) = \mathcal{D}_0. \tag{53}$$

Proof. Fix i and a. Two different full labels $u, u' \in \mathcal{U}_i$ with the same low part a have different high parts. Indeed $u = ab = u' = ab'$ would imply $u = u'$. Their high Walsh characters are therefore orthogonal. Hence

$$\mathbb{E}_\eta |B_{i,a}(\eta)|^2 = \sum_{\substack{u\in\mathcal{U}_i \\ u_{\leq z}=a}} |b_{i,u}|^2. \tag{54}$$

Summing over i and a gives (53). □

Theorem 4.2 (Robust face compression)**.** *For every $L \geq 1$,*

$$\mathbb{P}_\eta\left(\sup_{\sigma\in\{-1,1\}^{\{p\leq z\}}} Z_0(\sigma,\eta) > L\nu_z\mathcal{D}_0\right) \leq L^{-1}. \tag{55}$$

Consequently there exists a high–prime environment η for which

$$\sup_\sigma Z_0(\sigma,\eta) \leq \nu_z\mathcal{D}_0. \tag{56}$$

For each fixed small–prime assignment σ, one also has

$$\mathbb{E}_\eta Z_0(\sigma,\eta) \leq \nu_z\mathcal{D}_0. \tag{57}$$

Proof. The grouping (52) gives

$$S_i(\sigma,\eta) = \sum_{a\in\mathcal{A}_z} \chi_a(\sigma) B_{i,a}(\eta). \tag{58}$$

Cauchy–Schwarz, uniformly in σ, gives

$$|S_i(\sigma,\eta)|^2 \leq \nu_z \sum_{a\in\mathcal{A}_z} |B_{i,a}(\eta)|^2. \tag{59}$$

After summing over i,

$$\sup_\sigma Z_0(\sigma,\eta) \leq \nu_z W_z(\eta). \tag{60}$$

Markov's inequality and lemma 4.1 prove (55). The finite average of W_z equals $\mathcal{D}_0$, so some environment satisfies $W_z \leq \mathcal{D}_0$; this gives (56). Taking expectations in (60) proves (57). □

No limiting or compactness argument is involved in the existential statement.

The theorem includes a deterministic statement about the purely smooth Walsh sector. Averaging all high signs after fixing the small signs to -1 leaves exactly the kernels supported on primes at most z.

Corollary 4.3 (The smooth physical sector)**.** *One has*

$$0 \le \mathcal{D}_0 + \sum_{\substack{\kappa>1 \\ P^+(\kappa)\le z}} \mu(\kappa)C(\kappa) \le \nu_z \mathcal{D}_0. \tag{61}$$

Proof. The middle expression is $\mathbb{E}_\eta Z_0(-\mathbf{1}_{\le z}, \eta_{>z})$. It is nonnegative, and its upper bound is (57). $\square$

Remark 4.4 (Why there is no union bound)*.* The random variable W_z dominates all small–prime assignments at once. We therefore pay the actual projection count ν_z, not a probability union over $2^{\pi(z)}$ vertices. Conditional moment bounds for each fixed vertex would in any case require controlling collisions after the low coordinates are removed; the grouping above handles those collisions exactly.

5 The quantitative cutoff at the TPC endpoint

We now convert the abstract complexity ν_z into an explicit power of X. Only the label size (28) is used; no distribution theorem for the physical labels is assumed.

Lemma 5.1 (Rankin bound for projected labels)**.** *Let $Y = X^{\lambda+o(1)}$, let $z = (\log X)^A$ with fixed $A > 1$, and let $\mathcal{U}$ be any set of squarefree integers $u \le Y$. The number $\nu_z(\mathcal{U})$ of distinct z-smooth parts $u_{\le z}$ satisfies*

$$\nu_z(\mathcal{U}) \le X^{\lambda(1-1/A)+o(1)}. \tag{62}$$

Proof. Every projected part is a squarefree z-smooth integer $a \le Y$. For $0 < s < 1$, Rankin's device gives

$$\begin{aligned} \nu_z(\mathcal{U}) &\le \sum_{\substack{a\le Y \\ a \text{ squarefree} \\ P^+(a)\le z}} 1 \\ &\le Y^s \prod_{p\le z}(1+p^{-s}). \end{aligned} \tag{63}$$

Choose

$$s = 1 - \frac{1}{A} + \epsilon_X, \qquad \epsilon_X = (\log\log X)^{-1/2}. \tag{64}$$

For sufficiently large X, one has $0 < s < 1$. Moreover,

$$\begin{aligned} \log \prod_{p\le z}(1+p^{-s}) &\le \sum_{n\le z} n^{-s} \ll_A z^{1-s} \\ &= \log X \, \exp\Bigl(-A\sqrt{\log\log X}\Bigr) = o(\log X). \end{aligned} \tag{65}$$

The factor Y^s contributes

$$X^{\lambda(1-1/A)+o(1)}, \tag{66}$$

which proves the claim. $\square$

The full admissible range obtained from this argument is worth recording. Put

$$A_* := \left(1 - \frac{1}{400\lambda_0}\right)^{-1} = \frac{250196}{249671} = 1.00210277\ldots. \tag{67}$$

For every fixed $1 < A < A_*$,

$$\lambda_0(1 - 1/A) < \frac{1}{400}, \tag{68}$$

so $z = (\log X)^A$ can be absorbed by the endpoint budget.

Theorem 5.2 (Quantitative robust freezing)**.** *Let*

$$A = \frac{501}{500}, \qquad z_X = (\log X)^A, \qquad L_X = \exp\left(\frac{\log X}{\sqrt{\log\log X}}\right). \tag{69}$$

Then

$$\nu_{z_X} \le X^{62549/26302500 + o(1)}, \tag{70}$$

and

$$\frac{1}{400} - \frac{62549}{26302500} = \frac{12829}{105210000} > 0. \tag{71}$$

Consequently $L_X \nu_{z_X} \le K_B$ for all sufficiently large X, and

$$\mathbb{P}_{>z_X}\left(\sup_{\sigma_{p \le z_X}} Z_0(\sigma, \varepsilon_{>z_X}) > K_B \mathcal{D}_0\right) \le L_X^{-1}. \tag{72}$$

Proof. Apply lemma 5.1 with $\lambda = \lambda_0 = 62549/52500$. Since $1 - 1/A = 1/501$, the exponent is

$$\frac{\lambda_0}{501} = \frac{62549}{26302500}. \tag{73}$$

The exact subtraction in (71) leaves a fixed positive margin. Since $L_X = X^{o(1)}$, that margin implies $L_X \nu_{z_X} \le K_B$. Now apply theorem 4.2 with $L = L_X$. □

The failure probability is summable on dyadic scales. This gives a global-field formulation analogous to the one in TPC–43, but robust under low-prime overwriting.

Corollary 5.3 (Almost–sure dyadic overwriting)**.** *Let $X_r = 2^r$, and fix one deterministic inherited packet at each scale. Couple the packets using one global independent prime–sign field. For almost every field, all sufficiently large r have the following property: after keeping its signs at $p > z_{X_r}$, the signs at every $p \le z_{X_r}$ may be replaced arbitrarily and simultaneously, and every resulting equality energy is at most $K_B(X_r)\mathcal{D}_0(X_r)$.*

Proof. By (72), the r-th failure probability is at most

$$\exp\left(-\frac{r \log 2}{\sqrt{\log(r \log 2)}}\right) \tag{74}$$

for all sufficiently large r, and these probabilities are summable. The first Borel–Cantelli lemma, which needs no independence between scales, proves the result. □

The packet sequence must be prescribed. No uniformity over every possible packet sequence is asserted. Also, overwriting the low signs does not overwrite the high signs: the latter remain random in this corollary.

6 Exact restriction descent toward the Möbius corner

Conditional expectation deletes Walsh coefficients. Physical de–randomization does something different: setting a sign to -1 subtracts paired coefficients. This section records that operation exactly.

Let $p_1 < \cdots < p_R$ be as in (34) and put

$$P_r := \prod_{j=1}^{r} p_j, \qquad P_0 := 1. \tag{75}$$

For squarefree m with $(m, P_r) = 1$, define

$$C_r(m) := \sum_{a|P_r} \mu(a) C(am). \tag{76}$$

Recall that C is extended by zero and $C(1) = 0$. Throughout this section, every sum over m is restricted to squarefree positive integers. We also use the convention $p_0 = 1$, so restriction to coordinates $p \le p_0$ means no restriction.

Proposition 6.1 (Exact physical restriction)**.** *For every $0 \le r \le R$,*

$$F(-\mathbf{1}_{\le p_r}, \varepsilon_{>p_r}) = \sum_{\substack{m \ge 1 \\ (m,P_r)=1}} C_r(m) \chi_m(\varepsilon). \tag{77}$$

The coefficients satisfy

$$C_r(m) = C_{r-1}(m) - C_{r-1}(p_r m) \qquad (m, P_r) = 1. \tag{78}$$

The physical signed sum is invariant under the descent:

$$F(-\mathbf{1}) = \sum_{(m,P_r)=1} \mu(m) C_r(m), \tag{79}$$

and at the terminal stage

$$F(-\mathbf{1}) = C_R(1). \tag{80}$$

Proof. Write every squarefree kernel uniquely as $\kappa = am$, where $a \mid P_r$ and $(m, P_r) = 1$. At the restricted face, $\chi_a(-\mathbf{1}) = \mu(a)$, so grouping by m proves (77). Splitting the divisors of $P_r = P_{r-1} p_r$ according as they contain p_r gives (78). Evaluating the remaining signs at -1 gives (79). After every active prime has been processed, the only possible remaining index is $m = 1$, proving (80). □

Define the restriction energy

$$E_r := \sum_{(m,P_r)=1} |C_r(m)|^2. \tag{81}$$

Thus $E_0 = V$ and $E_R = |F(-\mathbf{1})|^2$.

Theorem 6.2 (Exact coefficient–correlation product)**.** *Whenever $E_{r-1} > 0$, define*

$$\theta_r := \frac{2 \operatorname{Re} \sum_{(m,P_r)=1} C_{r-1}(m) \overline{C_{r-1}(p_r m)}}{E_{r-1}}. \tag{82}$$

Then

$$-1 \leq \theta_r \leq 1, \qquad E_r = (1-\theta_r)E_{r-1}. \tag{83}$$

If an energy first becomes zero, set every later $\theta_r = 0$; every later energy is then zero. With this convention,

$$\boxed{|F(-\mathbf{1})|^2 = V \prod_{r=1}^{R}(1-\theta_r).} \tag{84}$$

Proof. Partition the indices in E_{r-1} into pairs $m, p_r m$, with $(m, P_r) = 1$. Then

$$E_{r-1} = \sum_{(m,P_r)=1} \left(|C_{r-1}(m)|^2 + |C_{r-1}(p_r m)|^2\right), \tag{85}$$

$$E_r = \sum_{(m,P_r)=1} |C_{r-1}(m) - C_{r-1}(p_r m)|^2. \tag{86}$$

Expanding the second line gives (83). The inequality $2|ab| \leq |a|^2 + |b|^2$ gives $|\theta_r| \leq 1$. Iteration from $E_0 = V$ to $E_R = |F(-\mathbf{1})|^2$ proves the product formula. □

The order of the prime restrictions can be changed. The intermediate θ_r then change, but their product is forced by the endpoints. Thus (84) is an identity along a chosen coordinate path, not an assertion that the local correlations are independent.

We next relate the small–prime projection count to the restriction energy. Let $\mathcal{K}_z$ be the set of low parts $\kappa_{\leq z}$ of nonzero pair kernels. Since every pair kernel is the symmetric difference of two row labels,

$$|\mathcal{K}_z| \leq \nu_z^2. \tag{87}$$

Proposition 6.3 (Energy cost of freezing the low primes)**.** *After every active prime $p \leq z$ has been restricted to -1, the remaining coefficient energy $E_z = E_{r(z)}$ satisfies*

$$E_z \leq \nu_z^2 V. \tag{88}$$

Proof. For a fixed high kernel m, the restricted coefficient is a signed sum

$$C_{\leq z}(m) = \sum_{a \in \mathcal{K}_z} \mu(a) C(am), \tag{89}$$

with absent coefficients interpreted as zero. Cauchy–Schwarz gives

$$|C_{\leq z}(m)|^2 \leq |\mathcal{K}_z| \sum_{a \in \mathcal{K}_z} |C(am)|^2. \tag{90}$$

Summing over m, using unique low–high factorization, and applying (87) proves the claim. □

Although the exponent ν_z^2 is natural for the squared coefficient spectrum, theorem 4.2 is stronger for the nonnegative energy itself: it returns to the unsquared row labels and pays only one factor ν_z.

7 Conditional face transport and likelihood drift

The coefficient descent tracks the centered spectrum. There is a second exact descent that preserves the nonnegativity of Z_0. For $0 \leq r \leq R$, define

$$A_r := \mathbb{E}_{\varepsilon_{p_1},\dots,\varepsilon_{p_r}} Z_0(\varepsilon_{\leq p_r}, -\mathbf{1}_{>p_r}). \tag{91}$$

Then

$$A_0 = Z_0(-\mathbf{1}), \qquad A_R = \mathcal{D}_0, \tag{92}$$

and the reverse Walsh formula gives

$$A_r = \mathcal{D}_0 + \sum_{\substack{\kappa>1 \\ P^-(\kappa)>p_r}} \mu(\kappa)C(\kappa). \tag{93}$$

For an arbitrary cutoff z, use (35) and write

$$A_z := A_{r(z)}, \qquad E_z := E_{r(z)}. \tag{94}$$

Proposition 7.1 (Exact least–prime increment)**.** *Put*

$$L_r := A_{r-1} - A_r. \tag{95}$$

Then

$$L_r = \sum_{P^-(\kappa)=p_r} \mu(\kappa)C(\kappa). \tag{96}$$

Moreover,

$$0 \le A_{r-1} \le 2A_r. \tag{97}$$

If $Z_0(-\mathbf{1}) > 0$*, all* $A_r > 0$*, and*

$$\delta_r := \frac{L_r}{A_r} \in [-1, 1] \tag{98}$$

satisfies

$$\boxed{\frac{Z_0(-\mathbf{1})}{\mathcal{D}_0} = \prod_{r=1}^{R}(1+\delta_r).} \tag{99}$$

Proof. Subtract (93) at adjacent indices to obtain the increment formula. Before averaging ε_{p_r}, let B_- and B_+ be the two nonnegative conditional averages obtained by setting that sign to -1 and $+1$, respectively. Then

$$A_{r-1} = B_-, \qquad A_r = \frac{B_- + B_+}{2}, \tag{100}$$

which proves (97) and $-1 \le \delta_r \le 1$. If some $A_r = 0$, then $A_{r-1} = 0$, so positive A_0 forces every $A_r > 0$. Finally $A_{r-1} = A_r(1+\delta_r)$; telescope from $r = R$ to $r = 1$. ☐

The increment also has an exact Gram interpretation. After averaging the coordinates $p_1, \ldots, p_{r-1}$, group row labels by their incidence pattern B on those coordinates. With the later signs fixed at -1, let $V_{B,0}$ and $V_{B,1}$ be the resulting coefficient vectors in the direct sum of the output spaces, according as p_r does not or does divide the label. Orthogonality in the averaged coordinates gives

$$A_{r-1} = \sum_B \|V_{B,0} - V_{B,1}\|^2, \tag{101}$$

$$A_r = \sum_B \big(\|V_{B,0}\|^2 + \|V_{B,1}\|^2\big), \tag{102}$$

$$L_r = -2\,\mathrm{Re} \sum_B \langle V_{B,0}, V_{B,1}\rangle. \tag{103}$$

Consequently, whenever $A_r > 0$, the drift $\delta_r = L_r/A_r$ is a normalized real cross inner product between the p_r-free and p_r-divisible blocks.

The product has a probability interpretation. Let U be uniform measure on the active prime cube and define the energy–biased measure

$$d\mathbb{Q}_Z := \frac{Z_0}{\mathcal{D}_0}\, dU. \tag{104}$$

For $0 \le r \le R$, let

$$\mathcal{E}_r := \{\varepsilon_{p_{r+1}} = \cdots = \varepsilon_{p_R} = -1\}. \tag{105}$$

Theorem 7.2 (Boundary likelihood chain)**.** *If* $Z_0(-\mathbf{1}) > 0$*, then*

$$1 + \delta_r = 2\mathbb{Q}_Z(\varepsilon_{p_r} = -1 \mid \mathcal{E}_r)\,, \tag{106}$$

and therefore

$$\frac{Z_0(-\mathbf{1})}{\mathcal{D}_0} = \prod_{r=1}^{R} 2\mathbb{Q}_Z\big(\varepsilon_{p_r} = -1 \mid \varepsilon_{p_{r+1}} = \cdots = \varepsilon_{p_R} = -1\big)\,. \tag{107}$$

Proof. The uniform probability of $\mathcal{E}_r$ is $2^{-(R-r)}$, and conditioning the energy on that face gives A_r. Hence

$$\mathbb{Q}_Z(\mathcal{E}_r) = 2^{-(R-r)}\frac{A_r}{\mathcal{D}_0}. \tag{108}$$

The event $\mathcal{E}_r \cap \{\varepsilon_{p_r} = -1\}$ is $\mathcal{E}_{r-1}$. Dividing the corresponding masses gives

$$\mathbb{Q}_Z(\varepsilon_{p_r} = -1 \mid \mathcal{E}_r) = \frac{A_{r-1}}{2A_r} = \frac{1+\delta_r}{2}. \tag{109}$$

The product formula follows. $\square$

If $Z_0(-\mathbf{1}) = 0$, every upper endpoint is automatic. Otherwise, for a target factor $K \ge 1$, exact endpoint selection is equivalent to

$$\sum_{r=1}^{R} \log(1+\delta_r) \le \log K. \tag{110}$$

This is why pointwise $o(1)$ bounds for δ_r, or even a small quadratic variation, need not suffice: a long sequence of tiny positive drifts may still accumulate.

8 A vanishing–variance rough obstruction

We now show that the high–prime condition left by the preceding sections cannot be supplied by a black–box variance, fixed–moment, or low–influence theorem. The example belongs to the same abstract class as (18): it is the squared norm of a Walsh sum with distinct squarefree labels.

Fix z, choose distinct primes $q_1, \dots, q_R > z$, and choose a squarefree u_0 coprime to all of them. For $\lambda > 0$, put

$$S(\varepsilon) := \chi_{u_0}(\varepsilon)\left(1 - \frac{\lambda}{R}\sum_{i=1}^{R} \chi_{q_i}(\varepsilon)\right), \qquad Z(\varepsilon) := |S(\varepsilon)|^2. \tag{111}$$

The distinct row labels are $u_0, u_0q_1, \dots, u_0q_R$.

Theorem 8.1 (Rough degree–two late collapse)**.** *The energy* (111) *has the exact expansion*

$$Z = 1 + \frac{\lambda^2}{R} - \frac{2\lambda}{R}\sum_{i=1}^{R}\chi_{q_i} + \frac{2\lambda^2}{R^2}\sum_{1\le i<j\le R}\chi_{q_iq_j}. \tag{112}$$

Thus every nonconstant kernel is z-rough and the Walsh degree is at most 2. Moreover,

$$D := \mathbb{E}Z = 1 + \frac{\lambda^2}{R}, \tag{113}$$

$$Z(-\mathbf{1}) = (1+\lambda)^2, \tag{114}$$

$$\operatorname{Var} Z = \frac{4\lambda^2}{R} + \frac{2\lambda^4(R-1)}{R^3}. \tag{115}$$

Take $\lambda = R^{1/4}$. Then

$$D \longrightarrow 1, \qquad \frac{\operatorname{Var} Z}{D^2} = O(R^{-1/2}) \longrightarrow 0, \qquad \frac{Z(-\mathbf{1})}{D} \sim R^{1/2} \longrightarrow \infty. \tag{116}$$

For every fixed $1 \le q < \infty$,

$$\left\|\frac{Z}{D} - 1\right\|_{L^q(U)} \longrightarrow 0. \tag{117}$$

If $d\mathbb{Q}_Z = (Z/D)dU$, then

$$\chi^2(\mathbb{Q}_Z\|U) = \frac{\operatorname{Var} Z}{D^2} \longrightarrow 0, \qquad \|\mathbb{Q}_Z - U\|_{\mathrm{TV}} \longrightarrow 0, \tag{118}$$

while the density at the distinguished point diverges as in (116)*.*

Proof. The factor χ_{u_0} has modulus one and disappears on squaring. Expanding the remaining real square, using $(\sum_i \chi_{q_i})^2 = R + 2\sum_{i<j}\chi_{q_iq_j}$, gives (112). Orthogonality gives the mean and variance, while evaluating every χ_{q_i} at -1 gives (114).

For $\lambda = R^{1/4}$, write

$$\frac{\lambda}{R}\sum_{i=1}^{R}\chi_{q_i} = R^{-1/4}\left(R^{-1/2}\sum_{i=1}^{R}\chi_{q_i}\right). \tag{119}$$

Khintchine's inequality bounds every fixed L^q norm of the parenthesized variable. Hence the left side tends to zero in every fixed L^q, and squaring proves (117). The chi–square identity is the definition of $\mathbb{Q}_Z$, and total variation is at most one half of its square root. □

Even the primewise face increments are individually negligible. Order the q_i arbitrarily and define A_t as in (91). Direct averaging gives

$$A_t = \left(1 + \lambda\frac{R-t}{R}\right)^2 + \frac{\lambda^2 t}{R^2}, \tag{120}$$

and

$$A_{t-1} - A_t = \frac{2\lambda}{R}\left(1 + \lambda\frac{R-t}{R}\right). \tag{121}$$

Consequently

$$0 \le \delta_t \le \frac{2\lambda}{R}, \qquad \sum_{t=1}^{R}\delta_t^2 \le \frac{4\lambda^2}{R}. \tag{122}$$

For $\lambda = R^{1/4}$, the maximum is $O(R^{-3/4})$ and the sum of squares is $O(R^{-1/2})$, yet

$$\prod_{t=1}^{R}(1+\delta_t) = \frac{(1+\lambda)^2}{1+\lambda^2/R} \sim R^{1/2}. \tag{123}$$

For completeness, the normalized Walsh influence of a prime q_i is also $O(R^{-1})$. Indeed

$$\operatorname{Inf}_{q_i}(F) := \sum_{q_i|\kappa} |C(\kappa)|^2 = \frac{4\lambda^2}{R^2} + \frac{4\lambda^4(R-1)}{R^4}, \tag{124}$$

and division by (115) gives the claim at $\lambda = R^{1/4}$.

Remark 8.2 (Scope of the counterexample). The example proves that the abstract nonnegative Walsh–Gram interface does not force good behavior at the all–minus corner. It does not show that the actual packet coefficient $C_X(\kappa)$ realizes this example, nor that the physical endpoint fails. Its role is to identify which additional arithmetic statement is indispensable.

9 Literature boundary

The prime–by–prime conditioning in sections 3 and 7 is the Walsh analogue of a finite reverse martingale. Its differences group the coefficient spectrum by the least prime factor. Hypercontractive moment bounds for Rademacher multiplicative functions, such as Probability Result 2.3 of Harper [8], control such polynomials in the annealed model. They do not, by themselves, control a prescribed vertex of the cube. The robust face theorem uses the additional Gram representation and injective equality labels, rather than a stronger generic moment inequality.

General low–influence principles also have a different quantifier. The invariance theorem of Mossel et al. [9, Theorem 3.18] is distributional under bounded–degree and low–influence hypotheses. Even when those hypotheses hold, the theorem describes typical product–space inputs, not the all–minus input. The obstruction in theorem 8.1 makes this distinction explicit: its degree is two and its normalized influences vanish, yet its specified corner diverges.

Small–bias spaces provide a finite derandomization of Walsh averages. Naor and Naor [10, Theorem 3.2] construct such spaces with a logarithmic randomness seed. Applied to $F(\varepsilon) = \sum C(\kappa)\chi_\kappa(\varepsilon)$, the direct conclusion has the form

$$|\mathbb{E}_{\varepsilon\in\mathcal{S}} F(\varepsilon)| \le \epsilon \sum_{\kappa} |C(\kappa)|. \tag{125}$$

It therefore needs an ℓ^1 coefficient estimate and locates some good sign vector; it does not identify the physical vector. Classical Riesz–product results, for example Peyrière [11, Theorem 1.2], concern singularity of limiting product measures and likewise do not bound a specified finite corner.

Strong theorems for unweighted rough numbers do not supply the missing signed estimate. Gorodetsky [6, Theorem 1.1] proves a variance asymptotic for the indicator of integers without small prime factors in averaged short intervals. This controls spatially averaged roughness, not

$$\sum_{\kappa} \mu(\kappa) C_X(\kappa) \tag{126}$$

for a signed, packet–dependent coefficient. Large–small prime duality [1] and recent effective least–prime identities such as Tenenbaum [12, Theorem 1.1] exploit special divisor or harmonic weights. No corresponding convolution identity has been proved for C_X.

Möbius orthogonality to fixed–complexity structured sequences is a second possible route. For instance, Green and Tao [7, Theorem 1.1] give arbitrary logarithmic saving against polynomial nilsequences of fixed complexity. The present $C_X(\kappa)$ is scale dependent and is not known to be a fixed–complexity nilsequence, a bounded sum of such sequences, or a member of an analogous class. We therefore cite that theorem as a route boundary, not as a black box that applies here.

The closest classical diagnosis is the sieve parity problem. The asymptotic sieve of Friedlander and Iwaniec [5, Theorem 1] obtains a prime conclusion only after an explicit bilinear axiom is added to local–density and remainder information. In a broad modern sieve framework, Ford and Maynard [4, Theorem 2.1] show that insufficient Type–II information allows sequences satisfying nontrivial Type–I and Type–II estimates while still failing to force primes. Their theorems do not apply literally to the generally signed C_X, but they support the route map: rough–kernel counting alone cannot control (126); a coefficient–specific bilinear estimate or an equally strong arithmetic substitute is needed.

To avoid an accidental application of an unrelated theorem, we record the current negative information. The coefficient C_X is real but generally signed, packet and mask dependent, and changes with X. It is not known to be multiplicative, nonnegative, low degree, uniformly low influence, ℓ^1-small, smooth, divisor–convolutional, a fixed nilsequence, or controlled in a Type–II norm. The results cited in this section diagnose possible inputs; none proves the remaining physical corner estimate.

10 The residual high–prime gate

The small–prime theorem and coefficient descent combine into a precise sufficient condition. Process every active prime $p \le z_X$ first, where $z_X = (\log X)^{501/500}$, and call the resulting coefficient energy E_{z_X}. If it is nonzero, continue through the high primes and define

$$\mathfrak{A}_{>z_X} := \frac{E_R}{E_{z_X}} = \prod_{p_r > z_X} (1-\theta_r). \tag{127}$$

This is the exact high–face point–evaluation amplification. It may depend on the chosen order of the high primes through the individual factors, but the ratio itself does not.

Theorem 10.1 (A sufficient high–prime correlation gate)**.** *If the actual TPC equality coefficients satisfy*

$$\mathfrak{A}_{>z_X} \le X^{o(1)}, \tag{128}$$

then

$$Z_0(-\mathbf{1}) \le K_B \mathcal{D}_0 \tag{129}$$

for all sufficiently large X. A sufficient local condition for (128) is

$$\sum_{p_r > z_X} (-\theta_r)_+ = o(\log X). \tag{130}$$

Proof. If $E_{z_X} = 0$, then $F(-\mathbf{1}) = 0$ and the result is immediate. Otherwise, proposition 6.3 and (127) give

$$|F(-\mathbf{1})| \le \nu_{z_X} V^{1/2} \mathfrak{A}_{>z_X}^{1/2}. \tag{131}$$

By (33), $V^{1/2} \le X^{o(1)} \mathcal{D}_0$, while theorem 5.2 gives

$$\nu_{z_X} \le X^{1/400 - 12829/105210000 + o(1)}. \tag{132}$$

The fixed power margin absorbs the factors $X^{o(1)}$, as well as the diagonal term $\mathcal{D}_0$, and proves the endpoint.

For $-1 \le \theta < 1$,

$$\log(1-\theta) \le (-\theta)_+. \tag{133}$$

If a factor has $\theta = 1$, the terminal energy vanishes. Otherwise summing (133) proves that (130) implies (128). □

The theorem is conditional, but the condition is now both exact and coefficient specific. It asks that high–prime restrictions not create a polynomial cumulative anti–correlation. A plausible route is to prove a bilinear estimate for the paired coefficient sum

$$\sum_m C_{r-1}(m)\overline{C_{r-1}(p_r m)} \tag{134}$$

uniformly after the preceding restrictions. Merely proving that few labels contain p_r gives, by Cauchy–Schwarz, a square–root density loss; over many primes that is generally not summable. The needed input must also exploit an angle, sign, oscillation, or genuine source–target bilinear structure.

The conditional face product gives an equivalent one–sided route. If $A_z = 0$, then repeated use of (97) gives $A_0 = Z_0(-\mathbf{1}) = 0$, so the endpoint is automatic. If $A_z > 0$ and one can prove for some cutoff z

$$\log \frac{A_z}{\mathcal{D}_0} + \sum_{p_r \le z} (\delta_r)_+ \le \left(\frac{1}{400} + o(1)\right) \log X, \tag{135}$$

then (110) closes the endpoint. By (103), each δ_r is a normalized real cross inner product between the p_r-free and p_r-divisible blocks. A bound of size $O(1/p_r)$ for its positive part would be summable with only a polylogarithmic loss. No such estimate is asserted here.

10.1 What has and has not moved

TPC–43 proved that almost every full prime–sign environment closes the equality packet. The present theorem proves more: after drawing only the high signs, a density–one set of such environments permits arbitrary adversarial overwriting of every prime up to $(\log X)^{501/500}$. This includes the physical values on a growing block and is therefore a strict partial de–randomization.

The unresolved step is still substantial. One must either prove (128), establish the face drift bound (135), derive a Type–II estimate for (134), or identify a fixed structured class for C_X covered by a deterministic Möbius orthogonality theorem. Failure of one route would be informative, but none may be replaced by a suggestion that the all–minus high corner is random.

10.2 Claim ledger

The unconditional results of this paper are:

(i) exact forward and reverse prime martingale layers;

(ii) uniform compression of every small–prime face at cost ν_z;

(iii) the Rankin bound and robust freezing through $(\log X)^{501/500}$;

(iv) exact coefficient restriction and correlation products;

(v) exact conditional likelihood transport; and

(vi) the vanishing–variance degree–two rough obstruction.

The high–prime amplification bound is a clearly labeled sufficient hypothesis, not a proved estimate for the physical coefficients. No result here proves the twin–prime conjecture, a prime–pair asymptotic, fixed–shift four–Möbius cancellation, or a general violation of sieve parity.

11 Exact finite certificate

The companion script

`experiments/tpc44_certificate.py`

uses only Python's standard library and exact integer, rational, and Gaussian–integer arithmetic. It performs no random sampling. The frozen finite regressions verify:

(i) the low–high label split and projected diagonal identity;

(ii) the pointwise robust face inequality for every finite sign assignment in several exhaustive Gram packets;

(iii) the smooth physical sector after high–prime averaging;

(iv) the exact forward and reverse Walsh layers and their Parseval ledgers;

(v) the coefficient recurrence, physical invariance, and product formula in theorem 6.2;

(vi) the kernel projection bound and low–restriction energy estimate;

(vii) the conditional face increments and likelihood ratios; and

(viii) every formula in the rough degree–two obstruction, for a finite family of R and λ.

The script also checks the rational endpoint ledger

$$\lambda_0 = \frac{62549}{52500}, \qquad \frac{\lambda_0}{501} = \frac{62549}{26302500}, \qquad \frac{1}{400} - \frac{\lambda_0}{501} = \frac{12829}{105210000} > 0, \tag{136}$$

and scans its own syntax tree to exclude floating–point literals, random imports, true division, and optimization–sensitive `assert` statements.

The frozen run performs 15,677 exact checks. Its canonical certificate digest is

`24b7eb3f19af7a2c8a016672020ee593578463857f1d7fe3dabedd7e59b57433`.

The certified source SHA–256 is

`1bc866f4e7981b3beea6fd4682fc702bef7cc28b949c192a628e1582d69a5da1`,

and the canonical JSON SHA–256 is

`bc0f83c2ba177f395a4c6b22a2d5c5bf1773249134149a4fbce4326bb2e7d7ea`.

Normal and optimized Python executions produce byte–identical JSON.

Finite computation certifies only finite algebra and combinatorics. It does not certify the asymptotic Rankin argument, maximal order of ω, Bonami–Beckner inequality, Borel–Cantelli lemma, inherited TPC diagonal bound, or any physical Möbius cancellation. In particular, the certificate is not numerical evidence for twin primes or for the unproved high–prime gate.

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